

Doping Stability Report -- Al in KCoO2 at 200 C

Generated: 2026-05-30 14:53 Host: KCoO2 Dopant: Al

Executive Summary

Al preferentially occupies the Co layer. The formation energy difference between the two layers is +6.309 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is $-0.00125 \text{ \AA}^2/\text{ps}$ (stable), compared to $-0.00402 \text{ \AA}^2/\text{ps}$ on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-4.999	+0	No	PASS	PASS	STABLE
K	+1.310	+2	Yes	PASS	PASS	METASTABLE

Al at the Co layer -- STABLE

Charge situation

Al replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	-4.9992 eV	PASS	< 1.0 eV to pass
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$E_f = -4.999 \text{ eV}$. This value is exceptionally large and negative, indicating that Al incorporation is strongly thermodynamically driven at the Co site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	-0.00125 $\text{\AA}^2/\text{ps}$	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.172 $\text{\AA}^2$	----	
Max displacement	0.737 $\text{\AA}$	----	
Coordination (min/mean/max)	4 / 4.1 / 5	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.792 $\text{\AA}$ (relaxed: 1.8)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = $-0.00125 \text{ \AA}^2/\text{ps}$. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = $0.172 \text{ \AA}^2$; maximum displacement from starting position = $0.74 \text{ \AA}$.

Coordination fluctuated between 4 and 5 (mean 4.1), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = $1.792 \text{ \AA}$.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Al is thermodynamically favorable and kinetically stable on the Co site at 200 C. All four stability criteria

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pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Al at the K layer -- METASTABLE

Charge situation

Al replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	+1.3102 eV	FAIL	< 1.0 eV to pass
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E_f = +1.310 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Al incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	-0.00402 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.059 Å²	----	
Max displacement	0.792 Å	----	
Coordination (min/mean/max)	4 / 4.0 / 4	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.798 Å (relaxed: 1.8)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00402 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.059 Å²; maximum displacement from starting position = 0.79 Å.

Coordination fluctuated between 4 and 4 (mean 4.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.798 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: METASTABLE

Al sits stably on the K site during MD at 200 C, but the formation energy exceeds the 1.0 eV equilibrium threshold. This configuration is metastable -- it may be achievable via non-equilibrium synthesis routes such as ion exchange or rapid quenching.

Site Preference Comparison

The Co site is preferred by 6.309 eV over the K site. Formation energy: -4.999 eV (Co) vs +1.310 eV (K).

Al preferentially occupies the Co layer. The formation energy difference between the two layers is +6.309 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00125 Å²/ps (stable), compared to -0.00402 Å²/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.